

Syrup-touch Functional Requirements Specification

- Document purpose:** describe the functionality implemented in the current Syrup-touch software so testers can validate the delivered behavior.
- System under test:** touchscreen syrup dispensing application shared by the ESP32-S2 target and the SDL2 Linux simulator.
- Source basis:** README, esp32s2/src/main.cpp, esp32s2/include/config.h, and repository recipe data as of 2026-04-13.

1. System Overview

The software controls a syrup dispensing machine with a 320x240 touchscreen user interface, 9 recipe slots, and 8 independent syrup pump channels. The same application logic runs on both the physical ESP32-S2 hardware and a Linux simulator through a hardware abstraction layer.

The tester-visible workflow is built around four application states:

- Main menu
- View recipe
- Select recipe for editing
- Edit dashboard

2. Operating Context

Area	Implemented behavior
User interface	Landscape touchscreen UI with a 3x3 numbered grid. Recipes are selected by slot number 1-9.
Dispensing channels	8 pump outputs mapped to syrup_1 through syrup_8.
Persistence	Recipe amounts are loaded from and saved to /syrup_mixes.json.
Execution targets	ESP32-S2 device build and Linux simulator build share the same business logic.

3. Functional Requirements

FR-1. Application startup

- On startup, the software shall initialize colors, load recipe data, draw the main menu, and log that the system is ready.
- If the recipe data file cannot be read, the software shall continue by using zeroed recipe amounts for all 9 recipes and all 8 syrup channels.

FR-2. Main menu presentation

- The software shall display a 3x3 grid of recipe buttons numbered 1 through 9.
- The software shall display a status bar at the top with the product label and the instruction text indicating that long press is used to view and pour.
- Touches inside the status bar area shall be ignored.
- Each grid tile shall provide visual pressed and released feedback when touched.

FR-3. Recipe viewing by long press

- A press of 300 ms or longer on a recipe tile shall open the view screen for that recipe slot.
- The software shall show all 8 syrup values for the selected recipe in two columns.
- The view screen shall provide a BACK button and a POUR button.
- The current UI identifies recipes by slot number only. Stored recipe names are not displayed in the active UI.

FR-4. Entry to edit mode by passcode

- A short tap on a main menu tile shall be treated as input to a 4-step secret passcode buffer.
- The implemented passcode sequence shall be 4, 7, 2, 5.
- When 4 consecutive short taps match the passcode, the software shall enter recipe selection mode for editing.
- If more than 4 short taps occur, the buffer shall behave as a rolling 4-entry window.
- Entering recipe view by long press shall clear the in-progress passcode buffer.

FR-5. Recipe selection for editing

- After passcode acceptance, the software shall show a 3x3 selection grid for choosing which recipe slot to edit.
- A tap on any recipe slot shall open the edit dashboard for that recipe.

FR-6. Recipe editing behavior

- The edit dashboard shall show syrup channels S1 through S8 and a SAVE button in the ninth tile.
- Tapping a syrup tile shall increment that syrup amount for the selected recipe.

- The implemented increment cycle shall be 0 to 10 and then wrap back to 0.
- The software shall provide a brief visual flash on the edited syrup tile before redrawing its value.
- The current implementation does not provide a cancel or back action from the edit dashboard. Exit is through SAVE only.

FR-7. Saving edited recipes

- Tapping SAVE shall write the current recipe dataset to persistent storage.
- After SAVE, the software shall return to the main menu.
- The software shall log the saved recipe number and the on or off state of each motor channel.

FR-8. Pour initiation

- From the recipe view screen, tapping POUR shall start the dispensing process for the selected recipe.
- Only syrup channels with values greater than zero shall be scheduled for dispensing.
- If all 8 syrup values are zero, the software shall not display the summary screen and shall return to the main menu after the pour attempt.

FR-9. Pour execution rules

- Each syrup unit shall correspond to 10.0 seconds of motor runtime.
- No more than 2 motors shall run at the same time.
- When two motors are started in sequence, the second start shall be delayed by 500 ms to reduce inrush current.
- Each motor shall remain active until its configured amount has completed based on the time-per-unit setting.
- When a motor completes, the software shall turn that motor off and continue with any remaining queued motors.
- At the end of a pour, the software shall turn off all motors as a safety action.

FR-10. Pour progress screen

- During dispensing, the software shall show a progress screen for the active recipe.
- The progress screen shall display overall percentage complete.
- The progress screen shall display completed and total volume in ml.
- The progress screen shall display elapsed time and estimated remaining time.
- The progress screen shall display up to 2 active motor progress indicators, matching the maximum concurrency limit.
- The progress screen shall provide STOP and PAUSE or RESUME actions.

FR-11. Pause and resume

- Tapping PAUSE shall stop all currently running motors without clearing the active pour.
- While paused, the screen title shall change to indicate paused state and the action button shall

change from PAUSE to RESUME.

- Tapping RESUME shall restart the motors that were active before the pause.
- Paused time shall be excluded from the reported elapsed pour time.

FR-12. Stop during pour

- Tapping STOP during dispensing shall cancel the current pour.
- The software shall allow STOP both during normal progress polling and during the 500 ms inrush stagger window.
- A stopped pour shall be treated as a completed interaction with stopped status rather than as an application fault.

FR-13. End-of-pour summary

- After a non-empty pour, the software shall show a summary screen.
- The summary shall indicate either POUR COMPLETE or POUR STOPPED.
- The summary shall display recipe number, elapsed time, number of motors used, and total configured volume in ml.
- The summary shall wait for an OK tap before returning to the main menu.

FR-14. Audible and log feedback

- The software shall issue a short touch tone for regular UI taps.
- The software shall play a three-tone completion signal after a completed pour.
- The software shall play a two-tone warning signal after a stopped pour.
- The software shall log major events including startup, heartbeat, recipe viewing, editing, motor start, motor completion, pause, resume, stop, and save operations.

FR-15. Periodic heartbeat

- While the application is running, the software shall emit a heartbeat approximately every 5 seconds.
- The heartbeat shall pulse the LED and log the current state and uptime.

4. Data Requirements

Item	Requirement
Recipe count	9 recipe slots indexed 1 to 9.
Syrup count per recipe	8 syrup values per recipe, mapped to motors 1 to 8.
Editable range	Each syrup value is editable in the range 0 to 10 inclusive.

Persistence format Recipe data is stored in a JSON-like text file written by the application.

5. Tester Focus Areas

- Verify the distinction between short tap and long press, using the implemented 300 ms threshold rather than the README wording.
- Verify that recipe names are not shown in the current UI even if name fields exist in stored data.
- Verify that edit mode cannot be exited without saving.
- Verify that empty recipes do not show a summary screen after POUR is pressed.
- Verify that pause excludes paused duration from the elapsed-time figure.
- Verify that STOP works both during active pouring and during the start stagger period.
- Verify that no more than 2 motors are active at once.
- Verify that all motors are off after completion, stop, and application exit from the pour loop.

6. Out of Scope for This Version

- No on-screen authentication beyond the hidden passcode sequence.
- No visible recipe-name management in the UI.
- No cancel-without-save path from edit dashboard.
- No per-motor calibration or per-syrup timing differences; all channels use the same seconds-per-unit setting.

This document describes current implemented behavior for testing. Where repository documentation and code differ, the implementation in the shared application logic was treated as the source of truth.